INFOSYS SPRINGBOARD INTERNSHIP

CLIPMIND AI: AI-POWERED VIDEO SUMMARIZATION AND KEY MOMENTS DETECTION PLATFORM

A PROJECT REPORT

Submitted by

GOTURI RUSHIKA REDDY

ABSTRACT

The rapid growth of long-form digital video has created a need for intelligent systems that can transform large amounts of multimedia content into concise and useful information. ClipMind AI is an AI-powered video summarization and key moments detection platform designed to automatically analyze uploaded videos, generate transcripts, create concise summaries, and identify important moments with timestamps.

The system integrates video processing, speech-to-text transcription, natural language processing, summarization, keyword extraction, and analytics into a single web-based platform. Users can upload videos and obtain transcripts, short and detailed summaries, key moments, highlights, and content insights without manually reviewing the entire video.

ClipMind AI also provides role-based access for Content Creators, Learners, Educators, and Administrators. Creators can manage and summarize their videos, learners can access transcripts and important sections, educators can convert lecture videos into learning resources, and administrators can manage users, content, analytics, and system activities.

The application is implemented using technologies including React, Vite, TypeScript, Node.js, Express, FastAPI, Python, SQLAlchemy, SQLite, FFmpeg, JWT authentication, and Docker, with AI-based processing used for transcription, summarization, and content analysis. The platform aims to reduce the time required to understand long-form video content while providing users with quick access to the most relevant information.

CHAPTER 1

INTRODUCTION

Overview

The rapid growth of digital video content has made it increasingly difficult for users to efficiently understand and consume long-form videos. Educational lectures, tutorials, meetings, presentations, interviews, and other informational videos may contain valuable information, but finding the most important content manually requires considerable time and effort.

ClipMind AI is an AI-powered video summarization and key moments detection platform developed to address this problem. The system allows users to upload videos and automatically processes the content to generate transcripts, concise summaries, detailed summaries, keywords, and important video moments with timestamps.

The platform combines video processing, speech-to-text technology, Natural Language Processing (NLP), Artificial Intelligence, and analytics into a single web application. Instead of watching an entire video to find relevant information, users can quickly understand the content through the generated summary and directly access important sections using timestamped key moments.

ClipMind AI also provides role-based access for different types of users:

- Content Creator – uploads and manages videos, generates transcripts and summaries, and views content analytics.
- Learner – accesses videos, transcripts, summaries, key moments, bookmarks, and learning history.
- Educator – converts lecture videos into summarized learning resources and monitors classroom content.
- Administrator – manages users, content, system activities, analytics, and platform settings.

The system is implemented using a modern full-stack architecture consisting of React, Vite, TypeScript, Node.js, Express, FastAPI, Python, SQLAlchemy, SQLite, FFmpeg, JWT authentication, and Docker. The architecture separates the frontend interface, backend services, database operations, and AI/video-processing workflows.

The main goal of ClipMind AI is to reduce the time and effort required to understand long-form video content by automatically extracting the most useful information and presenting it in an organized and accessible manner.

Advantages :

- *Saves Time – Users can understand long videos through automatically generated summaries instead of watching the entire video.*
- *Automatic Transcription – Converts spoken video content into searchable text.*
- *AI-Powered Summarization – Generates concise summaries from lengthy video content.*
- *Key Moments Detection – Identifies important sections and provides timestamps for quick access.*
- *Easy Content Discovery – Keywords, transcripts, and summaries make it easier to find relevant information.*
- *Role-Based Access – Provides different features for Content Creators, Learners, Educators, and Administrators.*
- *Centralized Platform – Video, transcript, summary, key moments, bookmarks, and analytics are available in one application.*
- *Improves Learning – Students and learners can quickly review important portions of lectures and educational videos.*
- *Content Analytics – Creators, educators, and administrators can monitor video and platform-related information.*
- *Docker Support – The application can be packaged and run consistently using Docker.*

Applications :

- Education and E-Learning – Summarizing lectures, tutorials, and online courses.
- Content Creation – Helping creators generate summaries and identify highlights from their videos.
- Corporate Training – Converting training sessions and presentations into concise learning material.
- Meetings and Conferences – Extracting important discussions and sections from recorded meetings.
- Research and Academic Work – Quickly reviewing long interviews, lectures, seminars, and recorded presentations.
- Media and Journalism – Finding important segments from interviews and long-form recordings.
- Online Learning Platforms – Providing transcripts, summaries, keywords, and important timestamps for learners.
- Knowledge Management – Converting large collections of recorded videos into searchable and structured information.
- Video Libraries – Helping users navigate large amounts of stored video content efficiently.
- Accessibility – Providing text-based transcripts and summaries that make video information easier to access.

CHAPTER 2

LITERATURE SURVEY

2.1 Paper Title: Automated Video Summarization Using Deep Learning

Author: Zhang et al. (2021)

Zhang et al. (2021) proposed an automated video summarization system using deep learning techniques to identify the most important information from lengthy video content. The system analyzes video frames and associated content to generate concise summaries that reduce the time required to watch complete videos. The study demonstrates that deep learning can effectively identify significant portions of videos and improve content accessibility. However, the system requires considerable computational resources for processing large video datasets and may face difficulties in accurately identifying important moments when the video contains complex or highly varied content.

2.2 Paper Title: Automatic Speech Recognition for Video Transcription

Author: Radford et al. (2022)

Radford et al. (2022) introduced Whisper, a large-scale automatic speech recognition system designed to transcribe speech from audio and video content. The model was trained on a large and diverse dataset and demonstrated strong performance across different languages, accents, and recording conditions. The study shows that automatic speech recognition can significantly reduce the effort required for manual transcription and can provide useful textual information for further processing. However, transcription accuracy may decrease when the audio contains heavy background noise, overlapping speakers, or unclear speech.

2.3 Paper Title: BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation

Author: Lewis et al. (2020)

Lewis et al. (2020) proposed BART, a sequence-to-sequence model designed for natural language generation and text transformation tasks. The model uses a denoising pre-training approach to understand and reconstruct corrupted text, making it effective for tasks such as text summarization. The study demonstrates that BART can generate fluent and meaningful summaries while preserving important information from the original text. However, the model requires significant computational resources and may occasionally omit important details when processing very long documents.

2.4 Paper Title: Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer

Author: Karpathy et al. (2014)

Karpathy et al. (2014) investigated the use of deep learning techniques for large-scale video classification and demonstrated how visual information from videos can be learned automatically using neural network architectures. The study highlights the ability of deep learning models to identify meaningful visual patterns from large collections of video data. However, processing video content requires substantial computational resources, and visual analysis alone may not be sufficient for understanding videos containing important spoken information.

2.5 Paper Title: Natural Language Processing for Automatic Text Summarization

Author: Nallapati et al. (2016)

Nallapati et al. (2016) proposed neural network-based approaches for abstractive text summarization, focusing on generating summaries that capture the important information from longer documents. The study demonstrated that neural sequence-to-sequence models can generate summaries that are more flexible than traditional extractive approaches. However, abstractive summarization may sometimes generate inaccurate information or omit important details, particularly when the input content is lengthy or complex.

2.7 Paper Title: AI-Based Video Summarization and Key Moment Detection

Author: ClipMind AI Project Specification (2026)

The ClipMind AI project proposes an integrated platform for automatically processing long-form video content through transcription, AI-powered summarization, keyword extraction, and key moments detection. The system is designed to reduce the time required to understand videos by providing concise summaries and timestamped important sections. It also supports role-based access for Content Creators, Learners, Educators, and Administrators. However, accurate key moment detection and high-quality summarization depend on the performance of the underlying AI models and the quality of the uploaded video and audio.

CHAPTER 3

PROBLEM DEFINITION

3.1 PROBLEM STATEMENT

With the rapid growth of online video content, users often need to spend a large amount of time watching complete videos to understand the important information. Educational lectures, tutorials, meetings, interviews, presentations, and other long-form videos may contain valuable information distributed throughout the recording.

Manual transcription, summarization, note-taking, and identification of important moments are time-consuming and require considerable human effort. Users may also find it difficult to locate a particular topic or important section quickly.

Therefore, there is a need for an intelligent platform that can automatically process video content, generate transcripts and summaries, identify important moments, and present the extracted information in an easy-to-use interface.

ClipMind AI is proposed to solve this problem by combining video processing, speech-to-text transcription, AI-based summarization, keyword extraction, key moments detection, and analytics in a single web-based platform.

3.2 EXISTING SYSTEM

In the existing system, users generally depend on conventional video players and separate tools to understand long-form video content. The user has to watch the complete video or manually search through the recording to find important information.

For transcription, users may depend on separate speech-to-text applications. Similarly, summarization, note-taking, bookmarking, and analytics may require different applications or manual effort.

The existing workflow can be summarized as:

Video → Manual Viewing → Manual Notes → Manual Summary → Manual Identification of Important Sections

The lack of an integrated system makes the process time-consuming and less efficient.

Common Features of Existing Systems:

- Manual video watching.
- Manual note-taking.
- Separate transcription tools.

- Manual summary preparation.
- Manual identification of important sections.
- Limited timestamp-based navigation.
- Separate tools for content analysis.
- Limited centralized analytics.
- No unified role-based video intelligence platform.

Limitations of the Existing System:

1. Time-Consuming – Users need to watch long videos completely to understand the content.
2. Manual Transcription – Converting spoken content into text manually requires significant effort.
3. Manual Summarization – Users have to prepare summaries themselves after watching the video.
4. Difficult Key Moment Identification – Important sections must be manually identified and recorded.
5. Multiple Tools Required – Users may need different applications for transcription, summarization, video playback, and note-taking.
6. Limited Searchability – Finding a specific topic within a long video can be difficult.
7. Limited Analytics – Conventional video systems may not provide detailed content and usage analytics.
8. No Integrated AI Processing – Traditional systems do not provide a unified workflow for transcription, summarization, and key-moment detection.
9. Higher Human Effort – Most content-processing activities depend on manual work.

3.3 PROPOSED SYSTEM

ClipMind AI is an AI-powered video summarization and key moments detection platform designed to automatically transform long-form video content into useful and structured information.

The system allows users to upload videos and automatically process them using AI and video-processing technologies. The uploaded video is analyzed to generate a transcript, concise summary, detailed summary, keywords, and important video moments with timestamps.

The proposed system also provides role-based access for:

- Content Creator
- Learner
- Educator
- Administrator

The overall workflow is:

Video Upload → Video Processing → Speech-to-Text → Transcript → AI Summarization → Key Moments Detection → Keywords & Analytics → User Dashboard

This provides users with a faster and more organized way to understand long-form video content.

3.5 KEY FEATURES OF PROPOSED SYSTEM

1.Video Upload

Users can upload videos to the platform for automatic processing and analysis.

2. Automatic Transcription

The system converts spoken audio from the video into text using speech-to-text technology.

3. AI-Powered Summarization

The platform generates concise summaries from the video transcript.

4. Detailed Summaries

Users can obtain more detailed information when a short summary is not sufficient.

5. Key Moments Detection

The system identifies important portions of the video and provides corresponding timestamps.

6. Keyword Extraction

Important keywords and topics can be extracted from the processed content.

7. Bookmarks

Users can save important summaries, highlights, or content for later reference.

8. Analytics Dashboard

The platform provides content and usage information through dashboards.

9. Role-Based Access

Different users receive features according to their roles and requirements.

10. Video Processing

FFmpeg and backend processing components support video and audio processing.

11. AI Processing

AI-based components support transcription, summarization, keyword extraction, and content analysis.

Advantages of the Proposed System:

- Reduces Video Consumption Time
- Users can understand the main content without watching the complete video.
- Automates Manual Tasks
- Transcription, summarization, and key-moment identification are automated.
- Improves Learning Efficiency
- Learners can quickly review summaries and important sections of lectures.
- Easy Access to Important Content
- Timestamped key moments allow users to directly access relevant portions.
- Centralized Platform
- Videos, transcripts, summaries, bookmarks, key moments, and analytics are available in one system.
- Supports Multiple User Roles
- Creators, learners, educators, and administrators can use role-specific features.

CHAPTER 4
SYSTEM STUDY

4.1 FEASIBILITY STUDY

System study is an essential phase in the development of any software project, as it involves analysing the existing system, identifying user requirements, and evaluating the practicality of the proposed solution. In the case of the ClipMind AI, the system study focuses on understanding the challenges involved in manually analysing long-form video content and determining how effectively the proposed system can address these issues.

The study ensures that the solution is efficient, reliable, and suitable for implementation in real-world environments. It also evaluates whether the system can be developed and deployed using available software, hardware, and technological resources while meeting user expectations. By analysing various feasibility aspects, the system study confirms that the proposed solution is practical, beneficial, and capable of reducing the time and effort required for understanding video content.

4.1.1 Legal Feasibility

Legal feasibility ensures that the system complies with applicable laws and regulations related to data usage, video content, software implementation, and user information. The proposed ClipMind AI system is designed to process user-uploaded video content and associated information such as transcripts, summaries, keywords, bookmarks, and user account details.

The system provides authentication mechanisms to restrict access to authorized users and helps protect stored application data. Video content processed through the system should be used only with appropriate permission from the content owner or according to applicable institutional and legal requirements. The application follows standard software development practices for handling user and video information.

Since the system is intended to provide video analysis and summarization services while maintaining controlled access to user information, it is legally feasible when used with properly authorized content and in accordance with applicable data and copyright requirements.

4.1.2 Operational Feasibility

Operational feasibility evaluates how effectively the system can be implemented and used in real-world environments. The proposed ClipMind AI system is designed with a user-friendly web interface that allows users to upload videos, view generated transcripts, access summaries, identify important moments, and manage their content easily.

The system simplifies the complex process of analysing long-form videos by automating transcription, summarization, keyword extraction, and key moment identification. Users can efficiently access important information without manually watching the complete video. Different user roles such as Content Creators, Learners, Educators, and Administrators are provided with features according to their requirements.

The system can therefore be easily adopted by users with basic computer knowledge and is operationally feasible for educational, content creation, training, research, and other video-based applications.

4.1.3 Technical Feasibility

Technical feasibility evaluates whether the required technologies and resources are available for developing and implementing the proposed system. ClipMind AI uses modern technologies such as React, Vite, TypeScript, Node.js, Express, Python, FastAPI, SQLAlchemy, SQLite, and FFmpeg for application development and video processing.

The system also uses AI-based components for speech-to-text transcription, summarization, and content analysis. Docker is used to package the application and its required dependencies into a consistent deployment environment.

The required software technologies are available and can be integrated to develop the proposed system successfully. Therefore, the ClipMind AI project is technically feasible and can be implemented using the available development environment and resources.

4.1.4 Economic Feasibility

Economic feasibility evaluates whether the proposed system can be developed and maintained within reasonable cost limits. ClipMind AI uses several open-source technologies and development frameworks, which reduces the overall development cost.

Technologies such as React, Node.js, Express, Python, FastAPI, SQLite, SQLAlchemy, FFmpeg, and Docker can be used without requiring expensive proprietary software licenses. The system also reduces manual effort involved in video transcription, summarization, and content analysis.

Since the proposed system can be developed using commonly available tools and resources while reducing the time and effort required for manual video analysis, the system is economically feasible.

4.1.5 Schedule Feasibility

Schedule feasibility determines whether the proposed system can be completed within the available development period. ClipMind AI is divided into different modules such as user authentication, video upload, video processing, transcription, summarization, key moment detection, bookmarks, analytics, and deployment.

The modular development approach allows each component to be implemented and tested systematically. The frontend and backend can also be developed and integrated independently before complete system testing.

Therefore, the proposed system can be developed, tested, and deployed within the planned project schedule, making it schedule feasible.

CHAPTER 5

SYSTEM REQUIREMENTS SPECIFICATION

System Requirements Specification defines the functional and non-functional requirements of the proposed ClipMind AIsystem. It describes the hardware, software, user, and system requirements necessary for developing and operating the application.

5.1 SOFTWARE REQUIREMENTS

Operating System	Windows, macOS, or Linux
Front End	React, Vite
Back End	Node.js, FastAPI
Database	SQLite, SQLAlchemy
Development Tools	Visual Studio Code, Docker
Hosting Services	Firebase, AWS, or Digital Ocean
APIs	FastAPI Swagger/OpenAPI
Libraries & Frameworks	Bootstrap/Tailwind CSS, Socket.IO, MQTT, Pandas, NumPy

5.2 HARDWARE REQUIREMENTS

Processor	Intel i5 or higher / AMD Ryzen 5 or higher
RAM	Minimum 8GB (16GB recommended)
Storage	Minimum 500GB SSD (1TB recommended)
Monitor	Full HD Display (1920×1080 resolution)
Peripherals	:Standard Keyboard, Mouse, Webcamv
Internet	Stable connection with at least 10 Mbps speed

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM ARCHITECTURE:

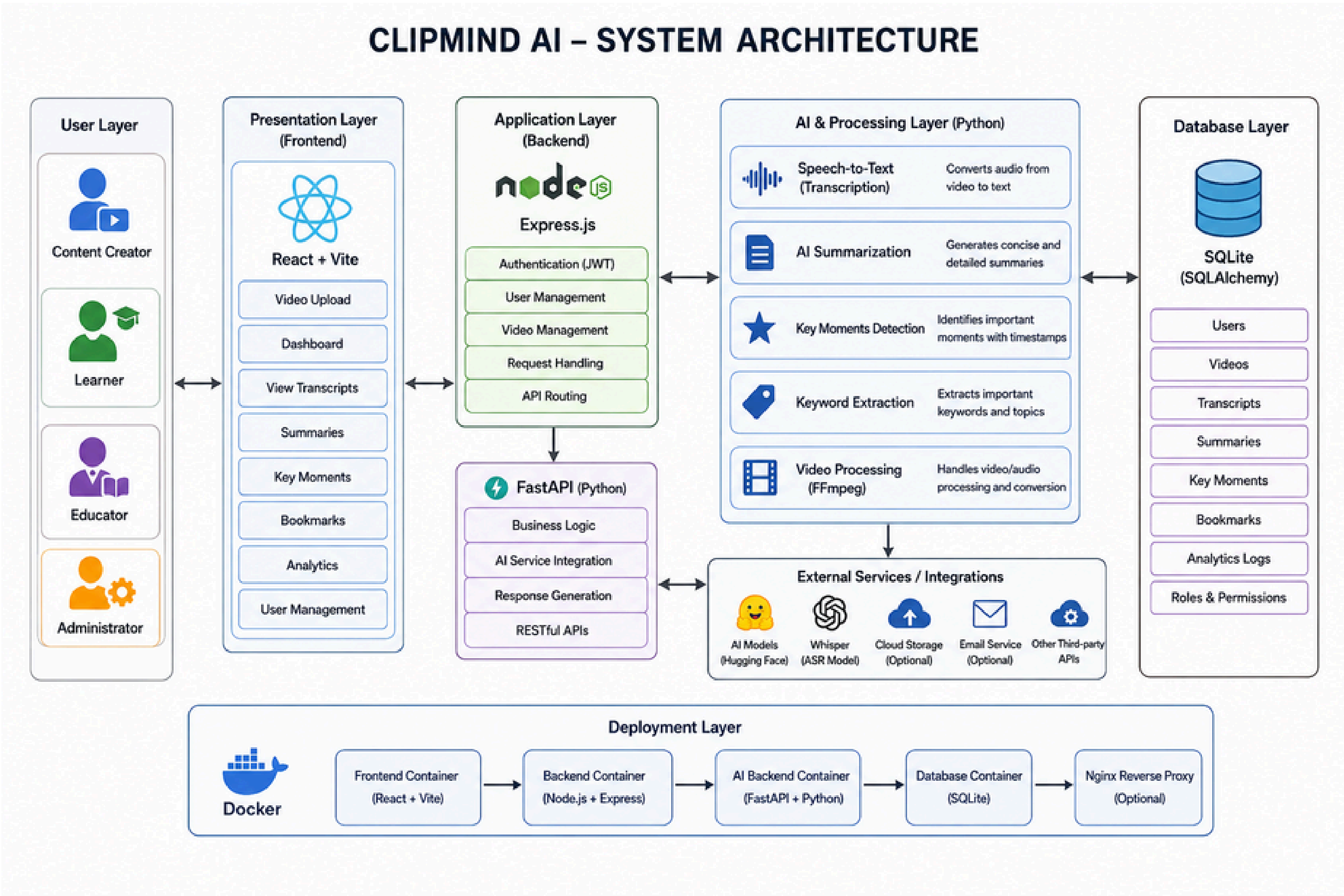


Fig 6.1: System Architecture

CHAPTER 7

SYSTEM TESTING AND MAINTANENCE

The purpose of system testing is to identify errors and ensure that the ClipMind AI system functions correctly. Testing is the process of executing the software to verify that it meets the specified requirements and user expectations without failures. The system is tested at various levels to ensure reliability, accuracy, and performance in handling video uploads, transcription, summarization, key moment detection, user management, and analytics.

TYPES OF TESTS

7.1 UNIT TESTING

Unit testing is performed to validate individual components of the ClipMind AI system. Each module is tested independently to verify that its functions produce the expected results.

The authentication module is tested for registration, login, password validation, and authorization. The video module is tested for video upload and management operations. The transcription and summarization modules are tested to ensure that the required content is generated correctly from the uploaded video.

These tests help identify errors at the individual module level before the components are integrated into the complete system.

7.2 FUNCTIONAL TEST

Functional testing is used to verify whether the system performs its specified functions correctly. The major functions tested include:

User registration and login, Role-based access, Video upload, Video processing, Transcript generation, AI-based summarization, Key moments detection, Keyword extraction, Bookmark management, Analytics, User management

7.3 INTEGRATION TESTING

Integration testing is performed to verify the communication between different modules of the ClipMind AI system.

For example, after a user uploads a video through the React frontend, the request is passed through the Express server to the FastAPI backend. The backend processes the video, generates the transcript and summary, stores the required information in the database, and returns the results to the frontend.

This testing ensures that the frontend, backend, AI processing components, and database work together correctly.

7.4 WHITE BOX TESTING

White box testing is used to examine the internal logic and structure of the system. It involves testing code-level operations such as authentication logic, data validation, API processing, video-processing operations, AI processing, and database queries.

This method helps identify errors within the internal workings of the application and ensures that logical conditions, processing operations, and database interactions are executed correctly.

7.5 BLACK BOX TESTING

Black box testing evaluates the ClipMind AI system from the user's perspective without requiring knowledge of its internal implementation.

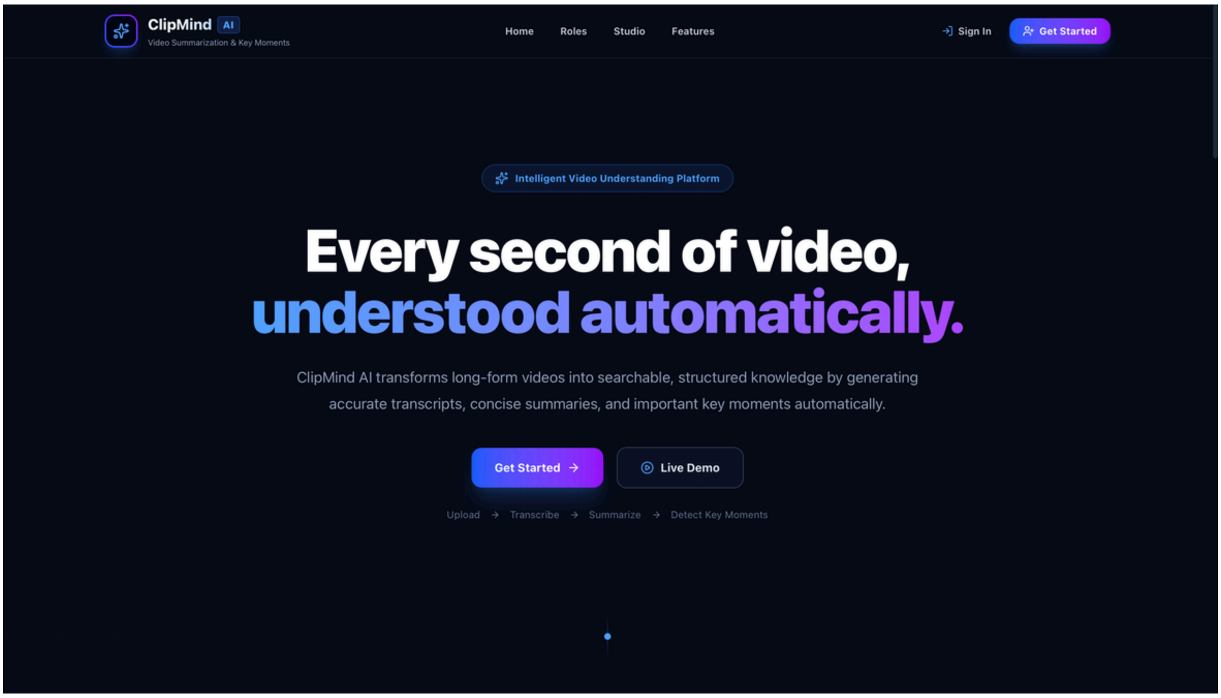
The testing focuses on whether the web interface loads correctly, input fields and buttons work properly, users can log in successfully, videos can be uploaded, and generated transcripts, summaries, keywords, and key moments are displayed correctly.

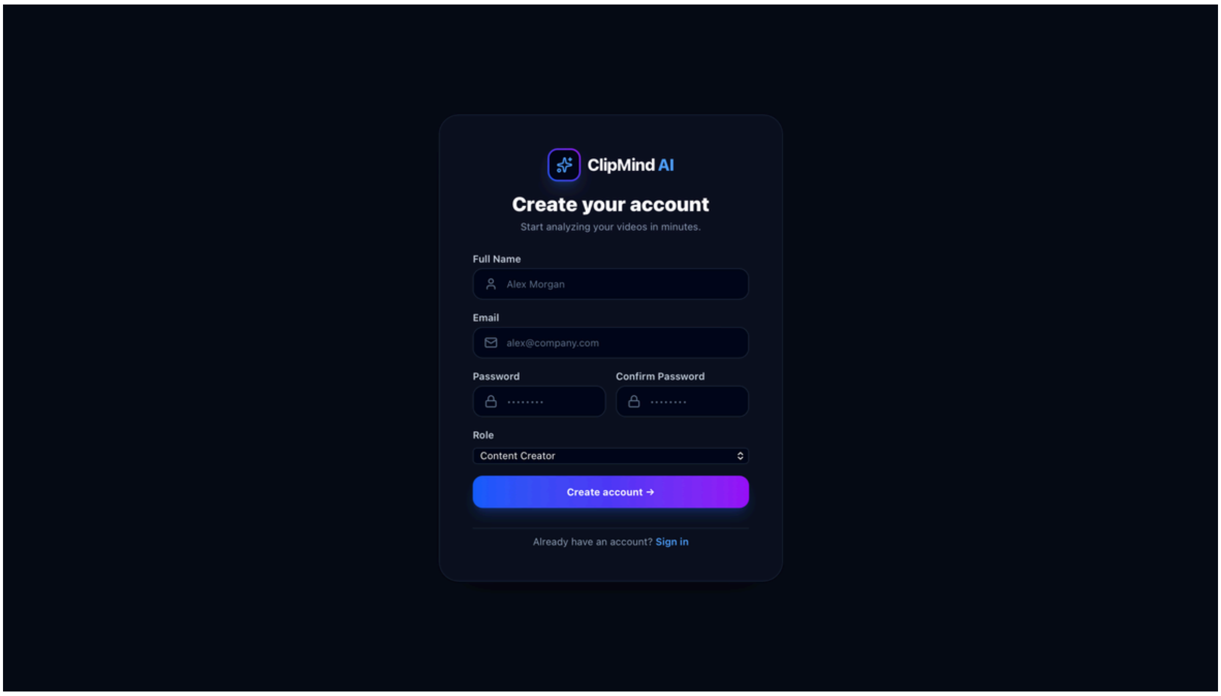
The system is also tested with invalid login credentials, unsupported inputs, and different video files to verify that appropriate responses and error messages are provided.

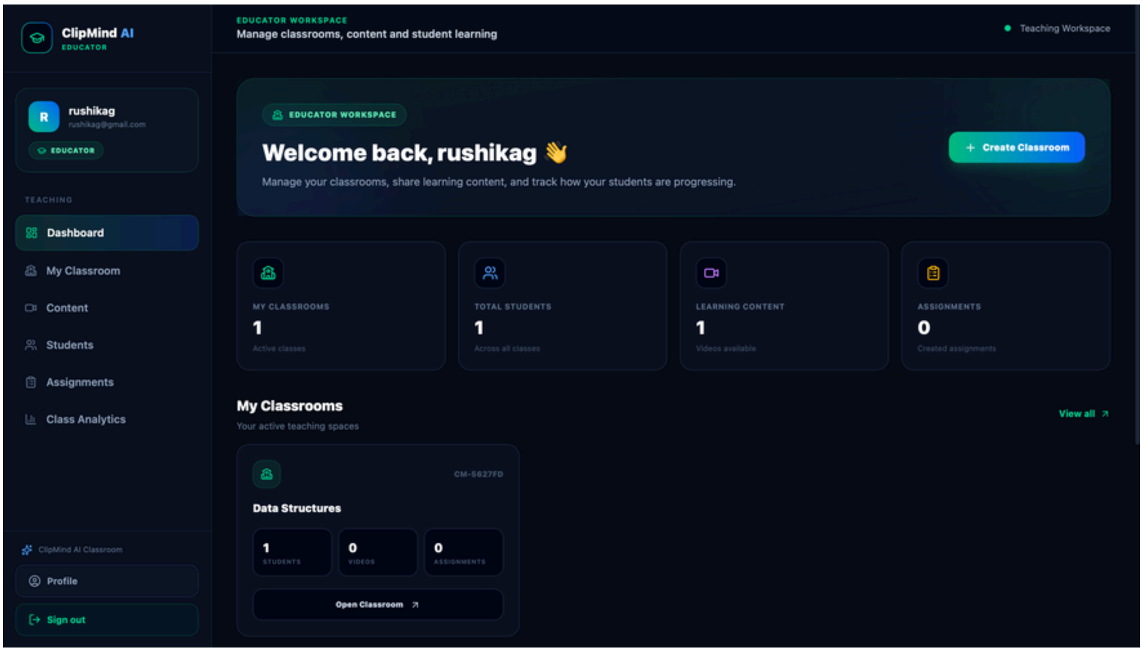
The primary objective of black box testing is to ensure that the application behaves correctly from the user's perspective and responds reliably without unexpected failures or crashes.

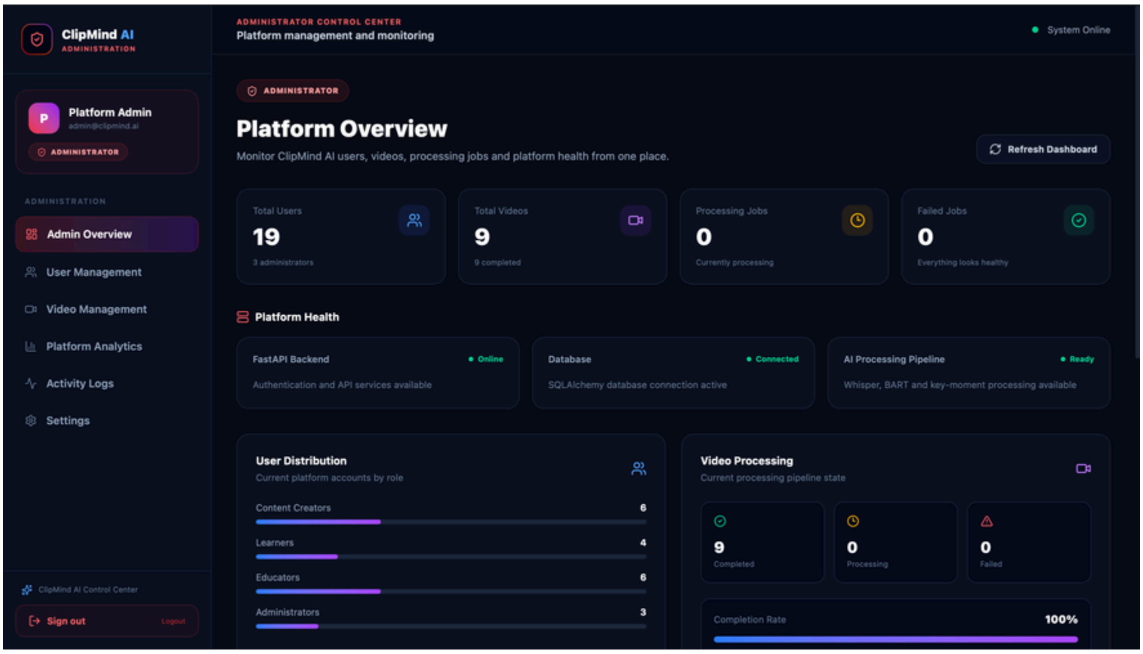
CHAPTER 8

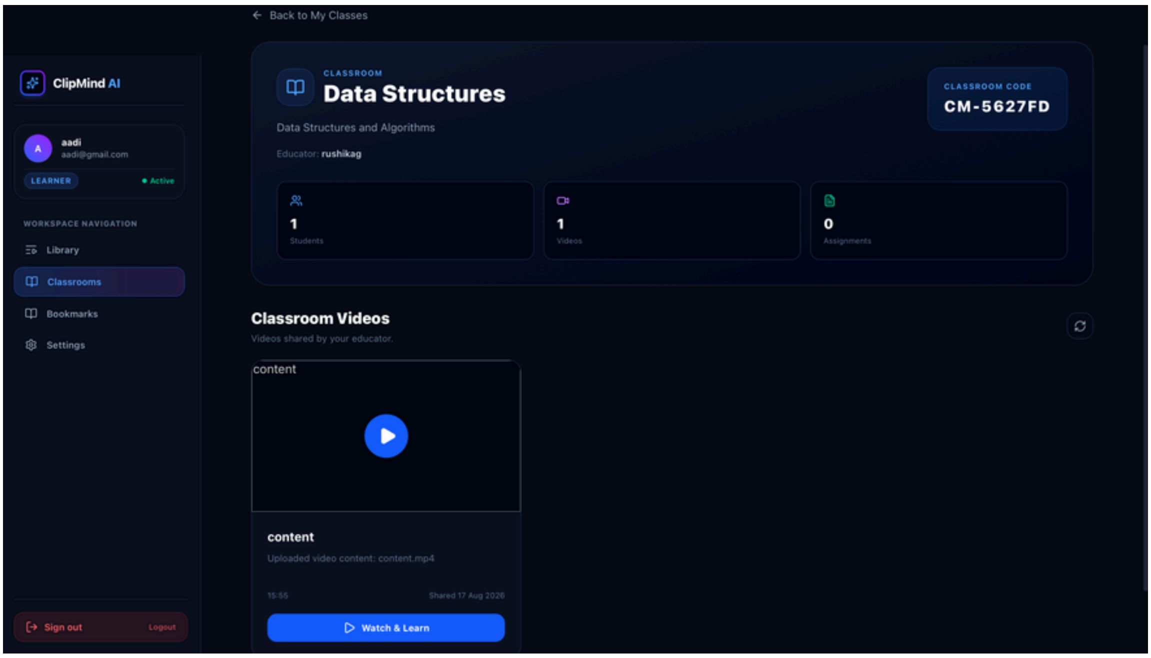
SCREENSHOTS






ClipMind AI
EDUCATOR
EDUCATOR WORKSPACE
Manage classrooms, content and student learning
Teaching Workspace
rushikag
rushikag@gmail.com
EDUCATOR
TEACHING
Dashboard
My Classroom
Content
Students
Assignments
Class Analytics
ClipMind AI Classroom
Profile
Sign out
EDUCATOR WORKSPACE
Welcome back, rushikag
Manage your classrooms, share learning content, and track how your students are progressing.
Create Classroom
MY CLASSROOMS
1
Active classes
TOTAL STUDENTS
1
Across all classes
LEARNING CONTENT
1
Videos available
ASSIGNMENTS
0
Created assignments
My Classrooms
Your active teaching spaces
View all
CM-5627FD
Data Structures
1
STUDENTS
0
VIDEOS
0
ASSIGNMENTS
Open Classroom


ClipMind AI
ADMINISTRATION
ADMINISTRATOR CONTROL CENTER
Platform management and monitoring
System Online
Platform Admin
admin@clipmind.ai
ADMINISTRATOR
ADMINISTRATION
Admin Overview
User Management
Video Management
Platform Analytics
Activity Logs
Settings
ClipMind AI Control Center
Sign out
Logout
ADMINISTRATOR
Platform Overview
Monitor ClipMind AI users, videos, processing jobs and platform health from one place.
Refresh Dashboard
Total Users
19
3 administrators
Total Videos
9
9 completed
Processing Jobs
0
Currently processing
Failed Jobs
0
Everything looks healthy
Platform Health
FastAPI Backend
Online
Authentication and API services available
Database
Connected
SQLAlchemy database connection active
AI Processing Pipeline
Ready
Whisper, BART and key-moment processing available
User Distribution
Current platform accounts by role
Content Creators
6
Learners
4
Educators
6
Administrators
3
Video Processing
Current processing pipeline state
9
Completed
0
Processing
0
Failed
Completion Rate
100%


Back to My Classes
ClipMind AI
aadi
aadi@gmail.com
LEARNER
Active
WORKSPACE NAVIGATION
Library
Classrooms
Bookmarks
Settings
Sign out
Logout
CLASSROOM
Data Structures
Data Structures and Algorithms
Educator: rushikag
CLASSROOM CODE
CM-5627FD
1
Students
1
Videos
0
Assignments
Classroom Videos
Videos shared by your educator.
content
content
Uploaded video content: content.mp4
15:55
Shared 17 Aug 2026
Watch & Learn

CHAPTER 9

CONCLUSION & FUTURE ENCHANCEMENT

11.1 CONCLUSION

The ClipMind AI system has been successfully developed to simplify and improve the process of understanding long-form video content. By integrating modern web technologies with AI-based video processing, the application enables users to upload videos and automatically obtain transcripts, summaries, keywords, and important moments with timestamps.

The system reduces the need for traditional manual methods such as watching complete videos, preparing notes, and manually identifying important sections. This helps save time, reduce manual effort, and improve the accessibility of useful information from lengthy video content.

Users can easily access processed video information through the user-friendly interface. The system also supports role-based access for different users and provides features such as video management, bookmarks, and analytics.

The integration of video processing, speech-to-text transcription, AI summarization, key moments detection, database management, and authentication makes ClipMind AI a practical solution for educational, content creation, training, research, and other video-based applications.

Overall, the project provides an efficient and intelligent approach to video content analysis and helps users quickly understand the most important information without having to watch the entire video.

11.2 FUTURE ENHANCEMENTS

The ClipMind AI system can be further enhanced with several advanced features to improve its functionality and usability. Integration of multi-language transcription and summarization can allow users to process videos in different languages and make the platform useful for a wider range of users.

The system can also be enhanced with more advanced AI-based key moment detection and content analysis to improve the accuracy of identifying important sections. AI models can be further optimized to provide more accurate summaries and better understanding of complex video content.

Additionally, the application can be enhanced with mobile application support, allowing users to upload videos, view summaries, transcripts, and key moments conveniently from smartphones and tablets.

Integration with Learning Management Systems (LMS) can also be introduced to allow educational institutions to connect summarized lecture content with existing learning platforms. Notification systems such as email or mobile alerts can inform users about completed video processing and newly generated content.